

Supplementary Information for "The networks of ingredient combinations as culinary fingerprints of world cuisines"

1 Supplementary Note 1 – Data Preprocessing and Representation

Our study utilizes CulinaryDB, a repository of recipes introduced in the main text, encompassing 23 distinct cuisines, each representing a geocultural area. Table S2 delineates the correspondence between cuisines and their respective geographical regions, providing essential context for comprehending the cultural origins and culinary traditions associated with each cuisine.

To prepare the raw culinary dataset for analysis, we performed a preprocessing step to standardize and simplify the ingredients set. Our preprocessing consisted in mapping groups of similar ingredients into single ingredients, thereby reducing the overall number of ingredients. We applied the preprocessing to the following cases:

- **Merging Redundant Entries:** Duplicate or near-duplicate ingredient entries referring to the same entity were merged into a single standardized form. This was essential to reduce noise introduced by inconsistent naming conventions across recipes.
- **Combining Ingredients of Similar Shape, Color, and Taste:** Different shapes of pasta and noodles were grouped together into the categories 'pasta' and 'noodles'.
- **Merging Fresh and Dry Forms:** Ingredients available in fresh, dry, and refrigerated forms were combined. For example, fresh peas, dry peas, and refrigerated peas were all grouped under 'peas'.
- **Generalizing Ingredient Subtypes:** Specific cultivars or commercial variations were merged into their general class. For instance, cherry tomatoes, plum tomatoes, and globe tomatoes were all grouped under 'tomatoes'.
- **Grouping by Transformative Process:** Ingredients that result from a crafting process (e.g., fermentation, baking, curing) were grouped into broader culinary categories. For example:
 - All types of cheese (e.g., mozzarella, cheddar, blue cheese) were grouped under 'cheese'.
 - All bread types made from cereal sources were grouped under 'bread'.
 - All fermented grape-based beverages (e.g., red wine, white wine, port wine) were merged as 'wine'.
 - All types of vinegar (e.g. balsamic, apple, rice, cider) were combined into 'vinegar'.
- **Grouping Animal Parts:** Ingredients from various parts of animals were all grouped under their respective animal categories. For example, chicken wings, chicken offal, chicken livers, chicken thighs and chicken breast were all combined into 'chicken'.
- **Other Examples:** All types of mushroom were merged into a single 'mushroom' entry.

This preprocessing was crucial to reduce the complexity of the dataset and to ensure a consistent representation of ingredients across recipes.

To provide additional context on recipe complexity across cuisines, we report for each cuisine the recipe in our dataset that contains the largest number of unique ingredients. Table S1 lists the recipe titles, the total number of ingredients, and the full ingredient lists. These recipes illustrate the variability in ingredient diversity and reflect characteristic culinary practices within each cuisine.

In Fig. 2 (b) and (d) of the main text, we presented heatmap representations of both the ingredient type popularity profile and the corresponding z-scores for all 23 cuisines. Additionally, for five of these cuisines, we depicted their popularity profiles and z-scores as bar charts (Fig. 2 (c) and (e)). Here, in Figs. S1 and S2, we extend this analysis to the remaining 18 cuisines.

2 Supplementary Note 2 – Food Type Graphs

In the main text, we have introduced food type graphs. By looking at the way ingredient types are combined, such graphs represent cuisines at a coarser level of organization. Figure S7 illustrates the three different graph representations of a cuisine: (i) bipartite recipe-ingredient network, (ii) ingredient projection network, (iii) ingredient type-ingredient type graphs. From (i) to (iii), each of these graphs correspond to a cuisine representation at an increasingly higher level of granularity.

Focusing on the coarsest granularity, we initially presented eight filtered food type graphs by retaining the 50 largest weighted edges to enhance visualization. Subsequently, we showcased the Maximum Spanning Trees (MSTs) for six food type graphs, providing a more distilled representation of food types connections. Here, we report the food type graphs for all the 23 cuisines. In Fig. S3 we show the backbones of the food type graphs, while in Fig. S4 we show the Maximum Spanning Trees of the unfiltered graphs.

Our examination of the food type graphs indicates varying levels of organization, ranging from single polarity to multi-polarity. Some graphs exhibit single polarity with a centralized structure, where a specific food category, such as in Indian and Mexican cuisines, serves as a central hub with numerous connections. In contrast, others demonstrate multi-polarity, featuring a more balanced structure with multiple central nodes. Examples of this balanced organization include French, Italian, and Japanese cuisines, where diverse food categories collectively contribute to the overall culinary identity.

Notably, we observe that the structure of each Maximum Spanning Tree is highly variable, suggesting that MSTs have the ability to highlight distinctive features of individual cuisines that may not be readily apparent from unfiltered food type graphs. However, we are still able to observe common patterns across the Maximum Spanning Trees. First, the Spices category emerges as a central node in the trees of numerous cuisines, hinting at its universal importance in defining culinary profiles. Additionally, Northern European cuisines consistently feature the Additive food type as the most central node across their respective Maximum Spanning Trees, suggesting a shared emphasis on this category within this culinary group.

To provide a geometrical picture of the proximity of food type graphs, we perform a 2D t-SNE analysis on the weighted adjacency matrices. Figure S5 illustrates the results for three different perplexity values. The projections appear compatible with the results of the hierarchical clustering performed in the main text, offering a consistent geometric representation of culinary relationships.

The 2D t-SNE unveils a culinary continuum, positioning Northern European and East Asian cuisines at opposite extremes, at all perplexity levels. This continuum reflects the inherent similarities and differences in the culinary characteristics of these regions, emphasizing the utility of t-SNE in capturing the geometric relationships between food type graphs.

To further demonstrate the efficacy of ingredient type graphs in capturing information about the geocultural regions they represent, we selected two food-related World Bank development indicators: Food Imports and Moderate to Severe Food Insecurity. Food Imports is defined as the percentage of merchandise imports comprising food items. Specifically, it includes commodities classified under the Standard International Trade Classification (SITC) sections 0 (food and live animals), 1 (beverages and tobacco), 4 (animal and vegetable oils and fats), and division 22 (oil seeds, oil nuts, and oil kernels). Moderate to Severe Food Insecurity refers to the percentage of the population experiencing moderate or severe limitations in accessing adequate food. This indicator is calculated as a three-year average to account for annual fluctuations. These indicators are defined at the country level. To aggregate them for our defined regions of S2, we employed weighted calculations: for Food Imports, we computed a weighted sum where the weights correspond to each country's merchandise imports. This approach ensures that countries with higher import volumes have a proportionate influence on the regional indicator. For Moderate to Severe Food Insecurity, we calculated a weighted sum using each countries populations as weights, thus accounting for the varying population sizes and providing a more accurate regional representation.

We find that both indicators positively correlate with the spectral radius of ingredient type graphs. This relationship is shown in S6, where each indicator is plotted against the spectral radius.

To analyze these relationships, we performed a robust linear regression fit, which down-weights the influence of outliers in the data. The best-fit line for each relationship is included in the plots. Given the relatively small sample size of our dataset, we use the Spearman Rank Correlation as the measure of correlation.

Table S1. Recipes with the Largest Number of Ingredients per Cuisine

Cuisine	Recipe Title	# Ingredients	Ingredient List
African	Moroccan Chili and 10,000 Grains of Sand	24	apricot; bay leaf; bread; butter; chicken; chili pepper; chive; coriander; cumin; ginger; lamb; lemon; mint; mustard; olive; paprika; parsley; pine; raisin; soy sauce; sugar; tomato; vinegar; water
American	Turkey Sweet Potato Shepherd's Pie and Cran-applesauce Sundaes	29	apple sauce; banana; butter; carrot; cayenne; celery; cheese; chicken; cranberry; cream; flour; garlic; ginger; marjoram; mustard; nutmeg; olive; orange; pea; pecan; rosemary; sage; salt; soy sauce; sugar; sweet potato; thyme; turkey; water
Australasian	Fish Chowder	21	allspice; bacon; bay laurel; butter; capsicum; cardamom; cayenne; chicken; cinnamon; clam; clove; cod; flour; ginger; mace; milk evaporated; mushroom; mustard; nutmeg; onion; potato
British Isles	A Scotsman's Shepherd Pie	21	butter; carrot; cinnamon; cream; egg; flour; garlic; ginger; lamb; mustard; olive; paprika; parsley; pea; potato; salt; soy sauce; sugar; tomato; vinegar; water
Canadian	Grey Cup Chili	26	bean; beef; carrot; celery; chili pepper; cilantro; cooking oil; coriander; corn; cumin; egg; garbanzo; garlic; kidney bean; lemon; lime; liquid smoke; mushroom; mustard; onion; onion red; oregano; sugar; vinegar; wheat; zucchini
Caribbean	Tortilla with Jerk Chicken	23	allspice; avocado; bean; canola oil; cayenne; cheese; chicken; cinnamon; cream; jalapeno; lime; nutmeg; orange; pepper; rice; sage; salt; scotch; soy sauce; sugar; thyme; vinegar; wheat
Central European	Venison with Chanterelles in Cream Sauce and Braised Red Cabbage	20	apple; brandy; bread; butter; caraway; carrot; celery; cream; current red; flour; leek; lemon; meat; mushroom; mustard; rosemary; sherry; soybean sauce; tomato; water
Chinese	Spicy Hotpot	20	bean; cabbage; cilantro; cooking oil; corn; ginger; lamb; lemon; liquid smoke; mustard; onion; orange; peanut oil; pepper; shrimp; soybean sauce; star anise; sugar; tomato; vegetable stock
East European	Bigos (Hunter's Stew)	22	bacon; basil; bay leaf; beef; cabbage; caraway; carrot; cayenne; cinnamon; flour; ginger; marjoram; mustard; paprika; pork; salt; sauerkraut; soy sauce; sugar; vinegar; water; wine
French	Porcini Pork Tenderloin	23	basil; bay laurel; brandy; butter; cream; fennel; garlic; honey; lavender; lemon; marjoram; mushroom; olive; oregano; parsley; pepper; pork; rosemary; shallot; tarragon; thyme; water; winter savory

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Table S1 – continued from previous page

Cuisine	Recipe Title	# Ingredients	Ingredient List
Greek	Ground Gyros Platter Deluxe	22	banana; bread; butter; cheese; chicken; coriander; cumin; flour; ginger; lamb; lemon; mustard; olive; onion; oregano; soy sauce; spinach; sugar; tomato; vinegar; water; yogurt
Indian	vegetable korma	31	asafoetida; bay laurel; buttermilk; capsicum; cardamom; cayenne; chili pepper; cinnamon; clove; coriander; cream; cucumber; cumin; curry leaf; french bean; garlic; ghee; ginger; ginger garlic paste; milk; onion; pea; pepper; poppy seed; potato; star anise; sunflower; tomato; turmeric; wheat; zucchini
Italian	Sauteed Baby Artichokes with Oven-Dried Tomatoes and Green-Olive Dressing	24	artichoke; basil; bay laurel; chive; cooking oil; cream; dill; egg; fennel; lavender; marjoram; mushroom; olive; onion; oregano; pepper; rosemary; salt; tarragon; thyme; tomato; vegetable broth; wine; winter savory
Japanese	Homemade Japanese Curry	22	allspice; apple; cardamom; carrot; chili pepper; clove; coriander; cumin; fennel; fenugreek; flour; mustard; nutmeg; pepper; potato; salt; star anise; sugar; tomato; turmeric; vinegar; water
Korean	Budae Jjigae (Korean Army Stew)	19	anchovy; baking powder; butter; chili pepper; cocktail; daikon; egg; fish; flour; kombu; pepper; pork; rice; scallion; sesame; soy sauce; tofu; watercress; wheat
Mexican	Epi's 50-Ingredient Super Bowl Nachos	27	artichoke; avocado; bacon; basil; buffalo; celery; chicken; chive; cilantro; cream; garlic; olive; onion red; orange; oregano; paprika; pork; potato; potato chip; salt; scallion; soybean sauce; spinach; tomato; tortilla chip; towel gourd; wheat
Middle Eastern	Couscous with Lamb Stew	21	bay leaf; butter; butternut pumpkin; carrot; cayenne; chickpea; chili pepper; habanero; lamb; olive; onion; oregano; paprika; pepper; rose; rosemary; saffron; thyme; turnip; water; zucchini
Misc. Portuguese	Kale and Cabbage Soup	22	basil; bay laurel; bay leaf; beef; butter; cabbage; celery; fennel; garlic; kale; kidney bean; lavender; marjoram; olive; oregano; parsley; rosemary; tarragon; thyme; tomato; water; winter savory
Scandinavian	Mum's Swedish Meatballs	20	allspice; bread; butter; cinnamon; egg; flour; garlic; ginger; milk evaporated; mustard; nutmeg; olive; parsley; pork; salt; sherry; soy sauce; sugar; vinegar; water
South American	Peruvian Menestron Soup	18	basil; bean lima; beef; cabbage; carrot; celery; cheese; corn; garbanzo; garlic; onion; pasta; pea; pepper; potato; spinach; tomato; vegetable oil

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Table S1 – continued from previous page

Cuisine	Recipe Title	# Ingredients	Ingredient List
South East Asian	Grilled Steak and Papaya Salad	25	beef; carrot; chive; cilantro; cooking oil; cream; dill; egg; fish; honey; lemon; lettuce; lime; mint; onion; papaya; parsley; peanut; peanut oil; salt; shallot; soy sauce; sugar; vinegar; watercress
Spanish	Seafood, Sausage and Bell Pepper Paella	25	bay leaf; cayenne; coriander; garlic; halibut; ham; lemon; lime; lobster; mussel; olive; onion; oregano; parsley; pea; pork; rice; saffron; salt; shrimp; thyme; tomato; vegetable oil; vinegar; wine
Thai	Thai Green Curry Chicken	26	anchovy; bacon; cinnamon; coconut; coconut milk; coriander; cornstarch; cranberry; cumin; curry powder; egg; fish; garlic; jasmine; lemon; lemongrass; lime; olive; onion; peanut; raisin; rosemary; salt; syrup; turmeric; vinegar

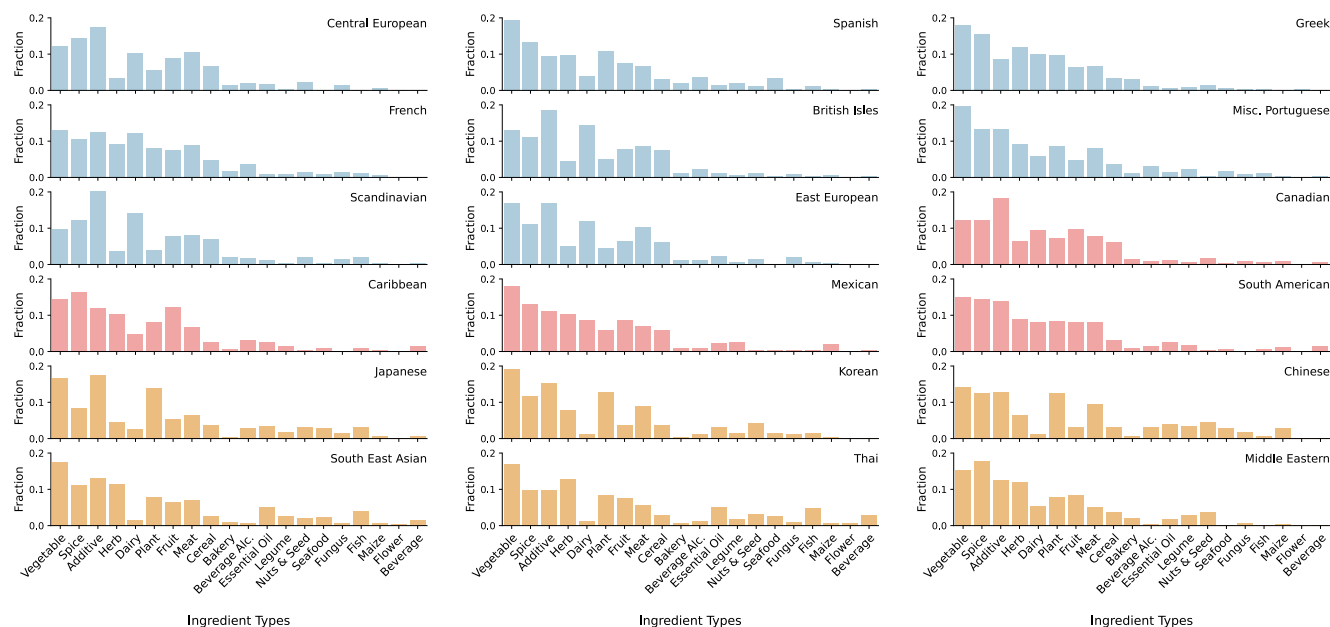


Figure S1. Ingredient Type Popularity Profile Bar chart representation of ingredient type popularity profiles for rest of the cuisines, highlighting the variations in ingredient preferences.

Table S2. Cuisine-Countries mapping

Cuisine	Countries covered
Indian	India, Pakistan, Bangladesh
African	Nigeria, Ethiopia, DR Congo, Tanzania, South Africa, Kenya, Uganda, Sudan, Algeria, Morocco, Angola, Ghana, Mozambique, Madagascar, Côte d'Ivoire, Cameroon, Niger, Mali, Burkina Faso, Malawi, Zambia, Chad, Somalia, Senegal, Zimbabwe, Guinea, Rwanda, Benin, Burundi, Tunisia, South Sudan, Togo, Sierra Leone, Libya, Congo, Central African Republic, Liberia, Mauritania, Eritrea, Gambia, Botswana, Namibia, Gabon, Lesotho, Guinea-Bissau, Equatorial Guinea, Mauritius, Eswatini, Djibouti, Comoros, Cabo Verde, Sao Tome & Principe, Seychelles
Chinese	China, Taiwan
Thai	Thailand
South East Asian	Cambodia, Laos, Vietnam, Philippines, Brunei, Malaysia, Singapore, Indonesia, Timor-Leste
Middle Eastern	Yemen, Oman, United Arab Emirates, Kuwait, Saudi Arabia, Jordan, Israel, Lebanon, Syria, Iraq, Iran, Turkiye, Armenia, Egypt, Qatar, Bahrain
Korean	South Korea, North Korea
Japanese	Japan
Australasian	Australia, New Zealand
Mexican	Mexico
Caribbean	Cuba, Jamaica, Haiti, Dominican Republic, Puerto Rico, Trinidad and Tobago
South American	Argentina, Bolivia, Brazil, Chile, Colombia, Ecuador, Guyana, Paraguay, Peru, Suriname, Uruguay, Venezuela
Central European	Germany, Austria, Switzerland
Greek	Greece
French	France
Misc Portuguese	Portugal
Italian	Italy
British Isles	England, Scotland, Wales, Ireland, Isle of Man, Inner and Outer Hebrides, Northern Isles
Scandinavian	Norway, Sweden, Finland, Aland Islands, Svalbard
East European	Poland, Latvia, Lithuania, Belarus, Ukraine, Moldova, Romania, Hungary, Slovakia, Czechia, Bulgaria, Russia
Spanish	Spain
American	United States of America
Canadian	Canada

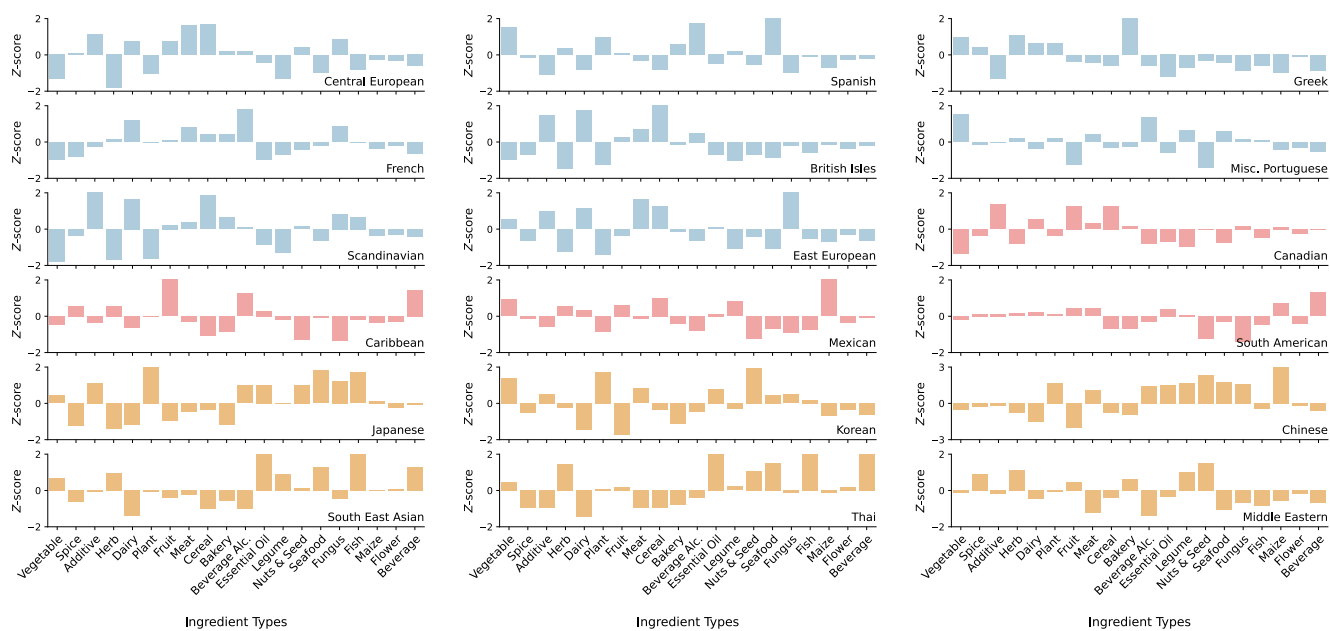


Figure S2. Z-score Profile Bar chart representation of z -scores for rest of the cuisines, revealing the relative significance of ingredient types.

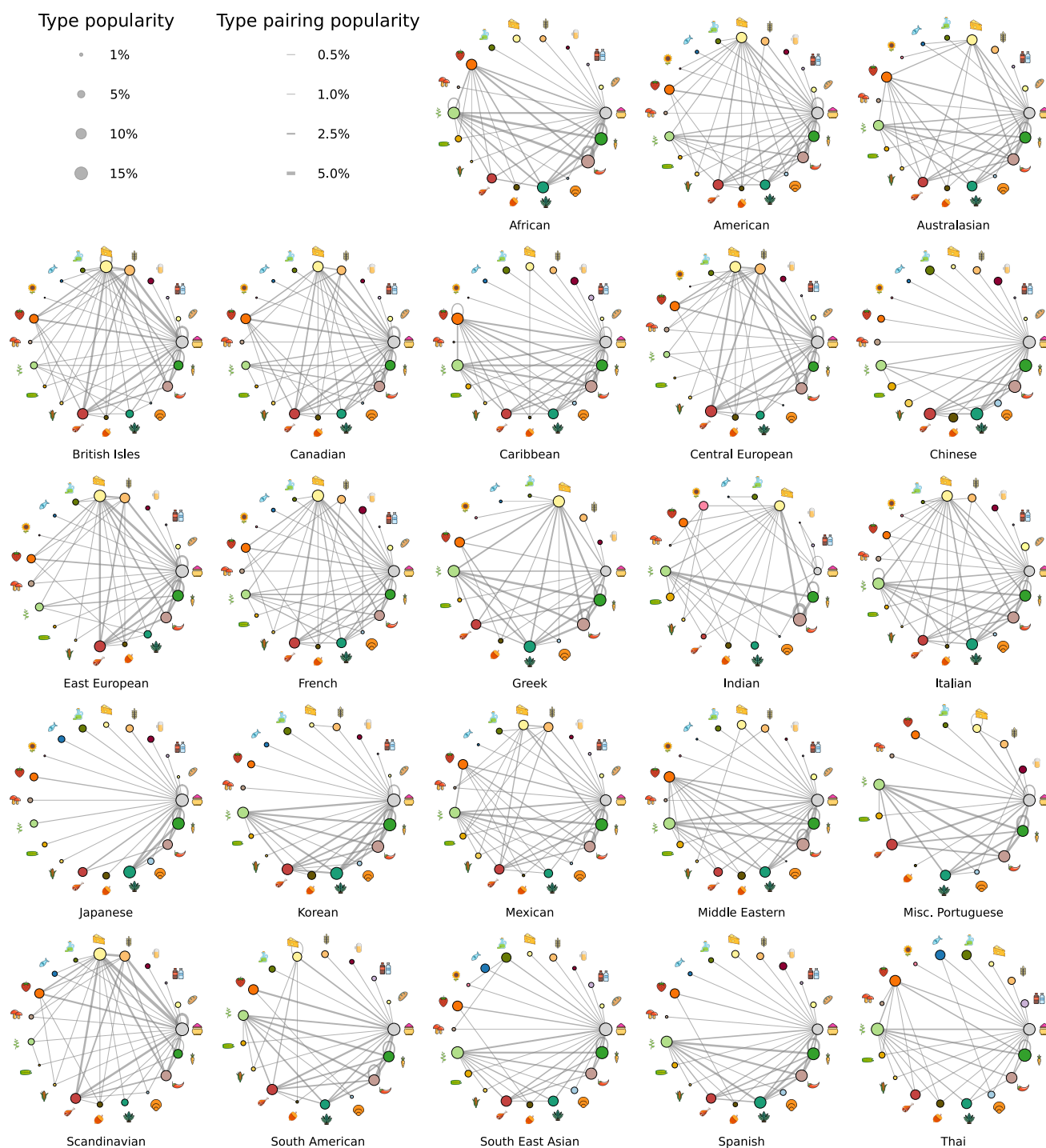


Figure S3. Networks of ingredient combinations. Ingredient combination networks for all 23 analyzed cuisines, filtered using the disparity filter method introduced by Serrano et al.⁴³. Refer to Fig. 3 (a-d) for details on the realization.

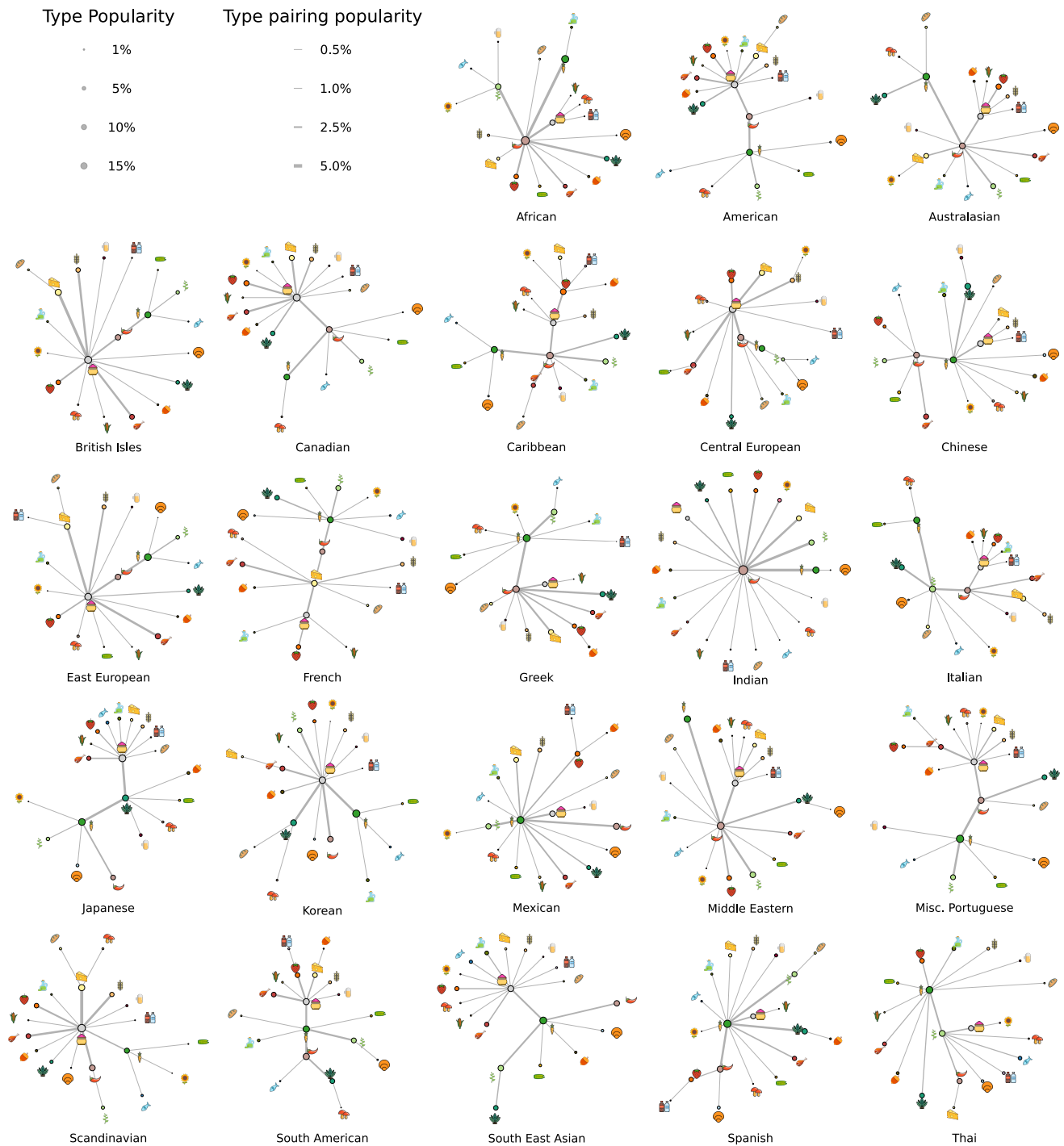


Figure S4. Maximum Spanning Trees of Ingredient Combination Networks. Maximum Spanning Trees of the Ingredient Type Graphs for all 23 cuisines. Details on the realization are reported in Fig. 4 of the main text.

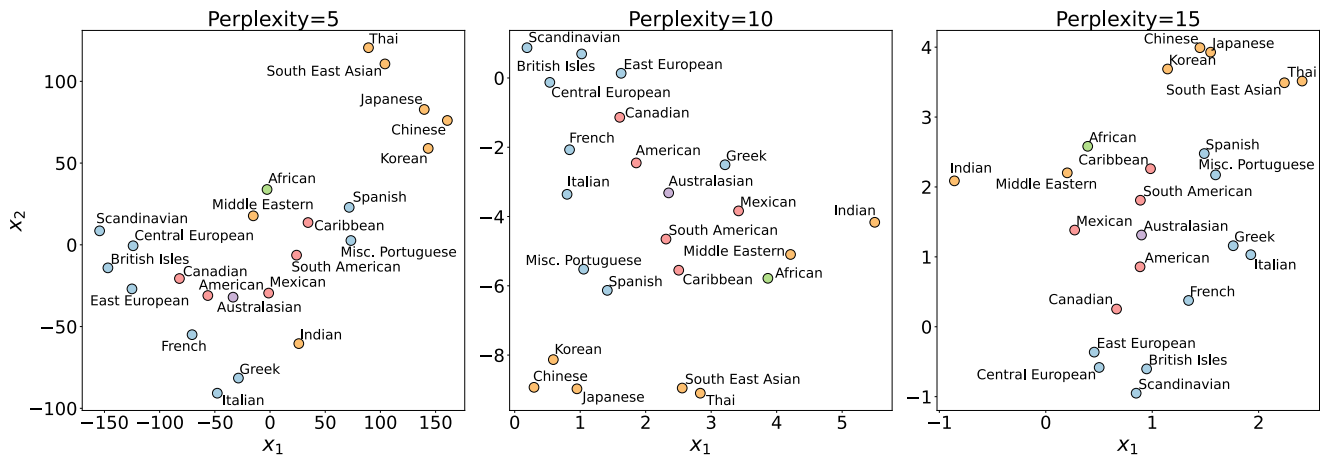


Figure S5. 2d t-SNE plots of ingredient type graphs. Each plot represents the 2 dimensional embedding of the weighted adjacency matrices of type graphs for all 23 cuisines. (a) Perplexity 5, (b) Perplexity 10, (c) Perplexity 15. Cuisines are colored according to their continent affiliation for additional context. Asian and Northern European cuisines exhibit clustering across all perplexity levels, indicating significant similarities in their ingredient usage.

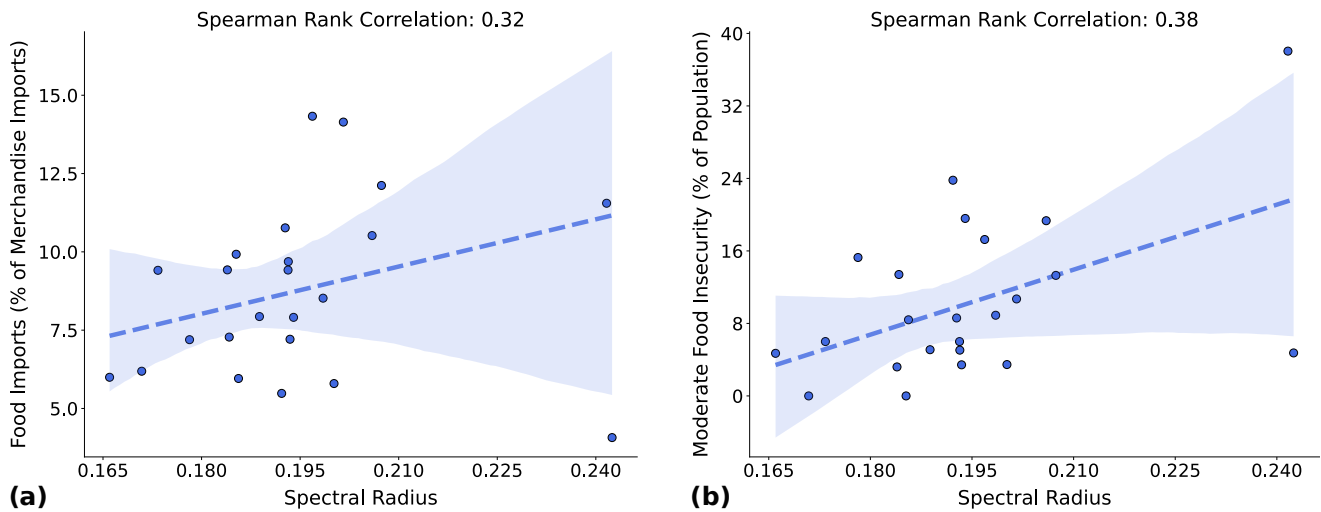


Figure S6. Scatter plot of World Development Indicators vs Spectral Radius for ingredient type graphs. (a) Relationship between the spectral radius of the ingredient type graphs (x-axis) and the percentage of Food Imports relative to merchandise imports (y-axis), for the geographic regions associated to each cuisine. (b) Relationship between the spectral radius (x-axis) and the percentage of Moderate to Severe Food Insecurity in the population (y-axis), for the same regions. Both plots show the best fit line obtained through a robust fit along with the 95% confidence interval as a blue shaded area. The Spearman rank correlation between the plotted variables is reported at the top of each plot.

Bipartite Representation of Recipes & Ingredients

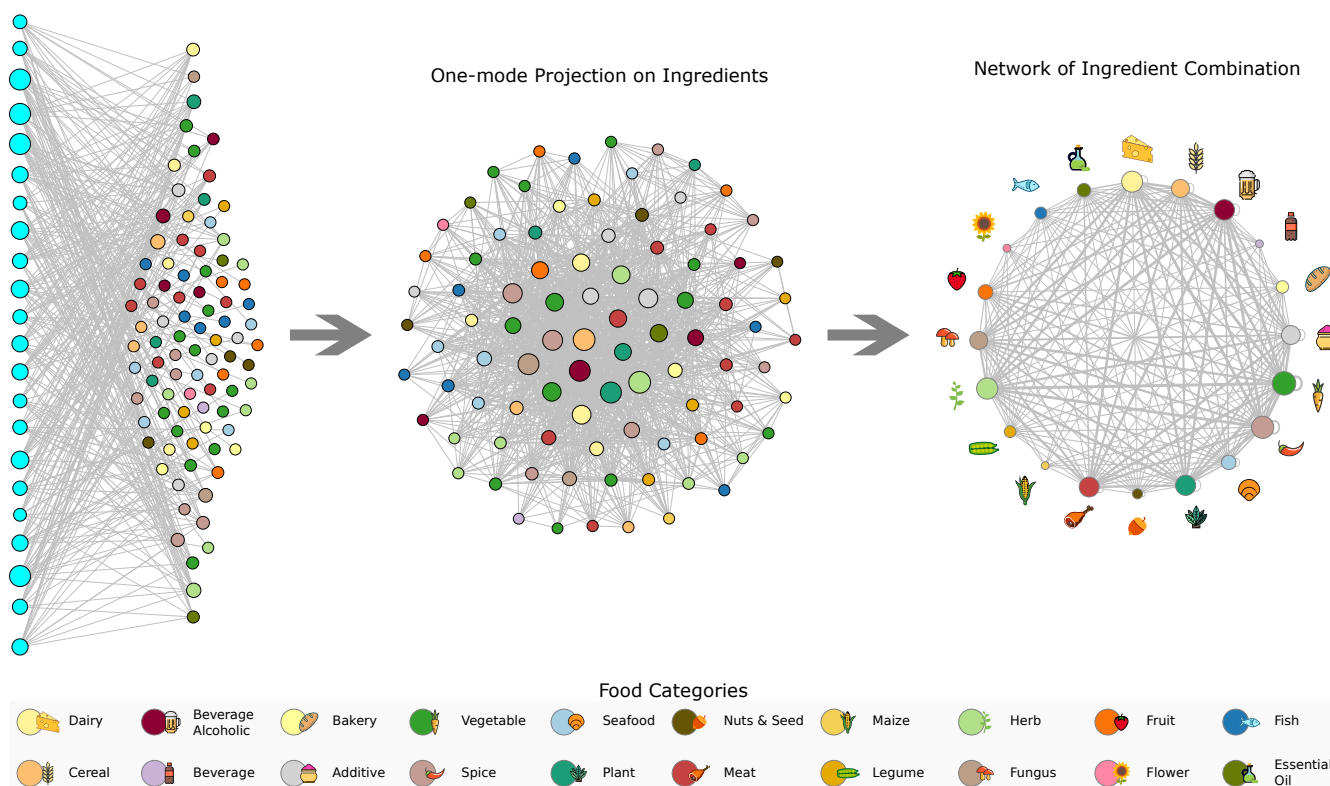


Figure S7. From bipartite representation to network of ingredient combination. The three different network representations of a cuisine (from left to right): (i) the bipartite recipe-ingredient network, where recipes and ingredients are connected if the ingredient appears in the recipe; (ii) the ingredient projection network, where ingredients are linked if they co-occur in at least one recipe; and (iii) the ingredient type-ingredient type network, where nodes represent ingredient categories and edges capture their co-occurrence across recipes. These representations capture the structure of a cuisine at progressively higher levels of granularity.